Questions

Q1.

A student investigates the motion of a trolley along a horizontal runway using the apparatus in Figure 4.

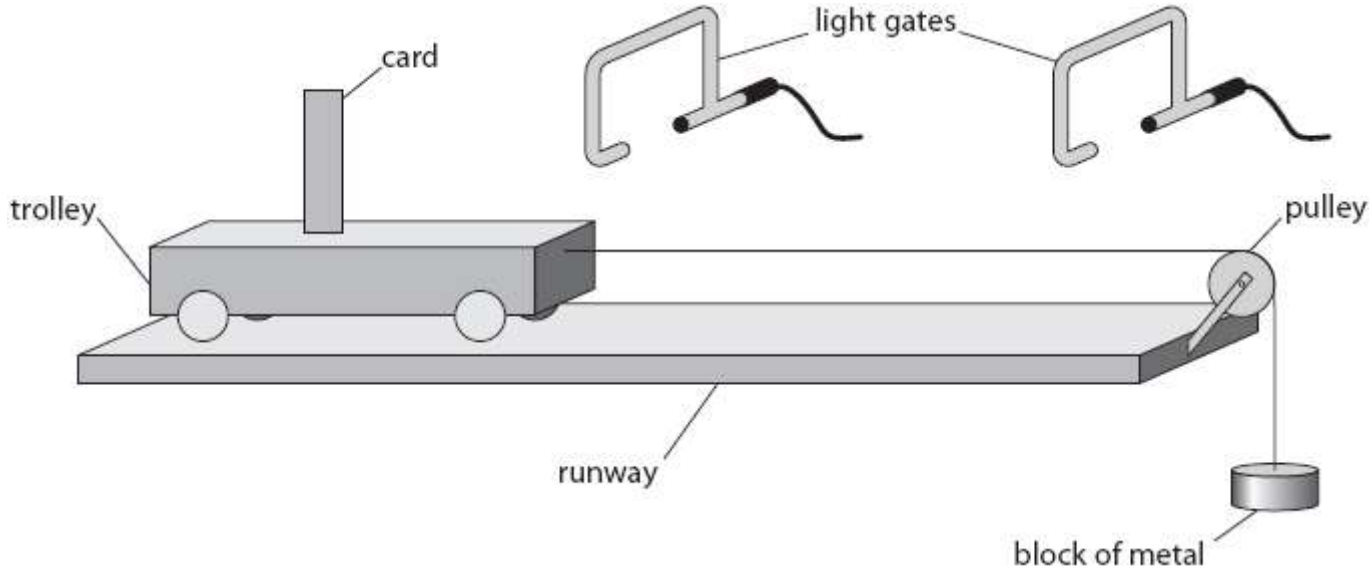


Figure 4

A trolley is attached to a string passing over a pulley.

A block of metal hangs on the end of the string.

Each light gate measures the time it takes for the card to pass through the gate.

When the trolley is released, it moves along the track.

A computer measures the time it takes for the card to pass between each light gate.

Figure 5 shows a graph of acceleration against force for three trollies of different mass that are pulled along the runway.

The graphs for the trollies are labelled P, Q and R.

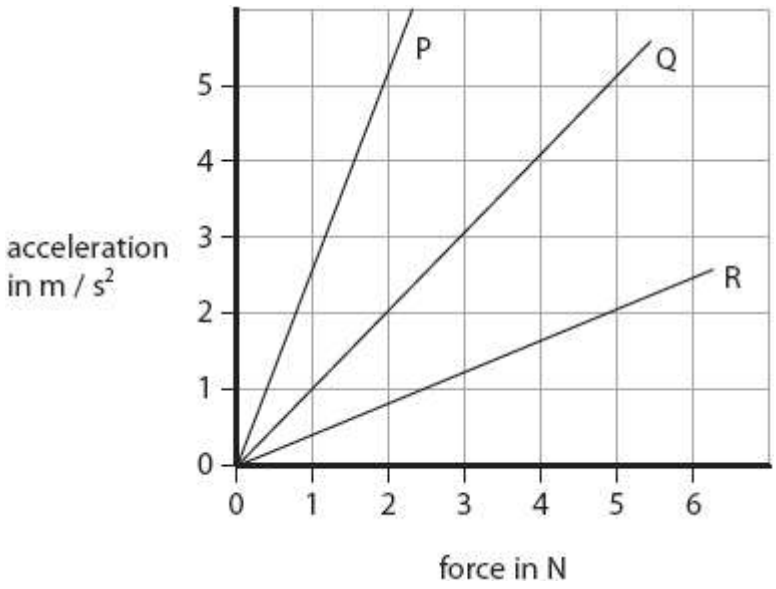


Figure 5

Use the information from the graph.

(i) Calculate the mass of trolley Q

(2)

mass of trolley Q = ... kg

(ii) Describe how the graph shows that trolley R has the greatest mass.

(2)

.....

.....

.....

.....

.....

(Total for question = 4 marks)

Q2.

A student investigates the motion of a trolley along a horizontal runway using the apparatus in Figure 4.

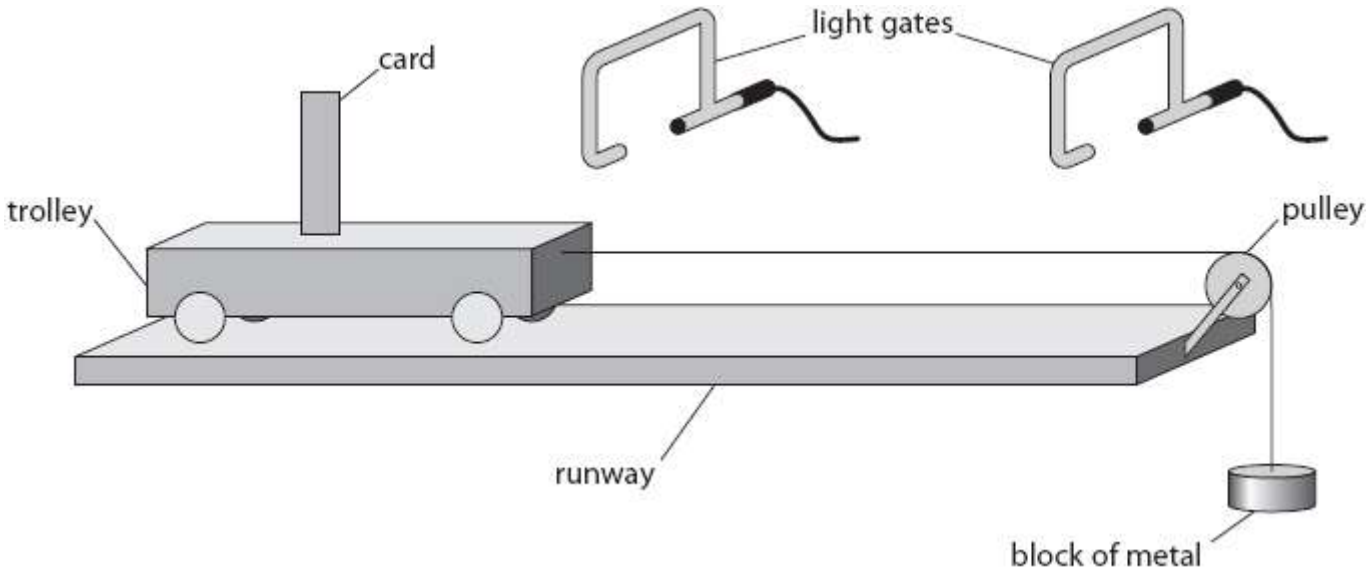


Figure 4

A trolley is attached to a string passing over a pulley.

A block of metal hangs on the end of the string.

Each light gate measures the time it takes for the card to pass through the gate.

When the trolley is released, it moves along the track.

A computer measures the time it takes for the card to pass between each light gate.

(i) The card took 0.080 s to pass through the first light gate.

The width of the card is 5 cm.
Calculate the average speed, in m/s, of the trolley through the first light gate.

(2)

average speed = ... m/s

Another trolley passes through the first light gate at a velocity of 0.72 m/s.

This trolley passes through the second light gate at a velocity of 1.1 m/s.

The time it takes for the card on the trolley to travel between the two light gates is 0.53 s.

(ii) State the equation relating acceleration, change in velocity and time.

(1)

(iii) Calculate the acceleration of the trolley between the two light gates.

(2)

acceleration = ... m/s^2

(Total for question = 5 marks)

Q3.

Figure 3 shows a Mars Exploration Rover.

(Source: *photojournal.jpl.nasa.gov*)

Figure 3

The mass of the rover is 190 kg.

(i) The gravitational field strength on Earth is 10 N / kg.

Calculate the weight of the rover on Earth.
Use the equation

weight = mass × gravitational field strength

(1)

weight on Earth = ... N

(ii) The weight of the rover on Mars is 700 N.

Calculate the gravitational field strength on Mars.

(2)

gravitational field strength on Mars = ... N/kg

(Total for question = 3 marks)

Q4.

The gravitational field strength on the Moon is 1.6 N/kg.
The mass of a rock on the Moon is 6.0 kg.

Calculate the weight of this rock on the Moon.

State the unit of weight.

Use the equation

weight = mass × gravitational field strength

(3)

weight of rock = ... unit .....

(Total for question = 3 marks)

Q5.

A rock on the surface of the Earth has a mass of 12 kg.

(i) Calculate the weight of this rock on the surface of the Earth.
The gravitational field strength on the surface of the Earth is 10 N/kg.
Use the equation

$$W = m \times g$$

(2)

weight on the Earth = ... N

(ii) The weight of the same rock on the surface of the Moon is 20 N.
Calculate the gravitational field strength on the surface of the Moon.

(3)

gravitational field strength on the Moon ..... N/kg

(Total for question = 5 marks)

Q6.

Quantities can be either scalar or vector.

The ball bearing is now dropped through air.

The initial velocity of the ball bearing is zero.

The acceleration of the ball bearing is 10 m/s^2.

The ball bearing falls 1.5 m.

Calculate the velocity of the ball bearing when it has fallen 1.5 m.

Use the equation

$$v^2 - u^2 = 2 \times a \times x$$

(2)

velocity of ball bearing = ... m/s

(Total for question = 2 marks)

Q7.

A motorcycle is travelling at a velocity of 6.2 m/s.

The motorcycle accelerates at 2.5 m/s^2 until its velocity is 10 m/s.

(i) Calculate the time taken for this acceleration.

Use the equation

$$\text{time taken} = \frac{\text{change in velocity}}{\text{acceleration}}$$

(2)

time taken = ... s

(ii) The motorcycle now decelerates (slows down) from 10 m/s to a stop.

The deceleration is at a constant rate of 4.4 m/s^2 .
Calculate the distance the motorcycle travels as it slows down to a stop.
Use the equation

$$v^2 - u^2 = 2 \times a \times x$$

distance = ... m

(Total for question = 2 marks)

Q8.

Figure 7 is a velocity-time graph for the motion of a van on a long, straight road.

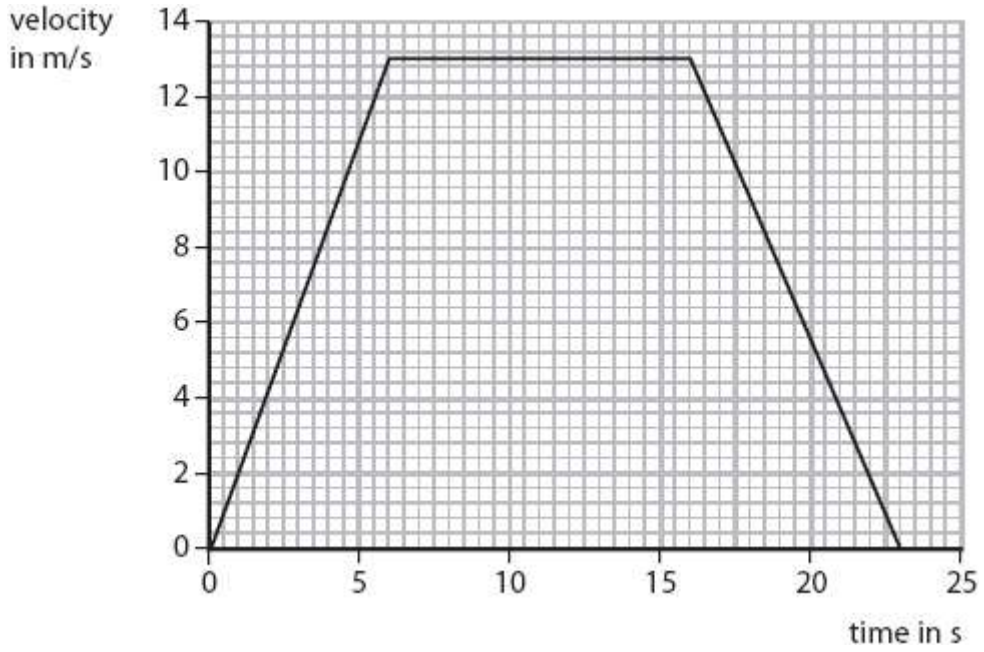


Figure 7

(i) The van accelerates at 2.2 m/s^2.

The force causing this acceleration is 5.5 kN.
Calculate the mass of the van.

(3)

mass = ... kg

(ii) Calculate the distance travelled by the van between 16.0 s and 23.0 s.
Give your answer correct to 2 significant figures.

(4)

distance travelled = ... m

(Total for question = 7 marks)

Q9.

A car with a mass of 1800 kg is accelerating at 1.2 m/s^2.

Calculate the force used to accelerate the car.

Use the equation

force = mass × acceleration

(2)

force = ... N

(Total for question = 2 marks)

Q10.

A car has a mass of 1200 kg.

Calculate the force needed to give this car an acceleration of 2.4 m / s^2 .

(2)

Use the equation

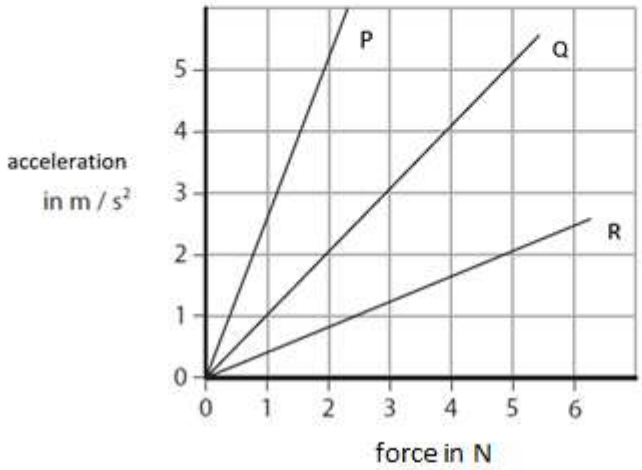
$F = m \times a$

force = ... N

(Total for question = 2 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	 any correct points from line Q e.g. substitution (1) mass = 2/2 evaluation (1) 1 (kg)	Accept answers in the range 1 to 1.05 (kg) full marks will be awarded for correct numerical answer without working	(2)

Question number	Answer	Additional guidance	Mark
(ii)	an answer that combines points of interpretation to provide a logical description: • (trolley/it) has the smallest acceleration for the largest force • (and) m = F/a		(2)

Q2.

Question number	Answer	Additional guidance	Mark
(i)	substitution and conversion (1) v = 0.05/0.08 evaluation (1) v = 0.63 (m/s)	full marks will be awarded for correct numerical answer without working	(2)

Question number	Answer	Additional guidance	Mark
(ii)	a= (v-u) / t (1)		(1)

Question number	Answer	Additional guidance	Mark
(iii)	substitution (1) (1.1-0.72) / 0.53 evaluation (1) 0.72 m/s^2	 full marks will be awarded for correct numerical answer without working	(2)

Q3.

Question number	Answer	Additional guidance	Mark
(i)	1900 (N)	allow 1862 1864 190 × 10 190 × 9.8(1)	(1)

Question number	Answer	Additional guidance	Mark
(ii)	rearrangement (1) $\frac{700}{190}$ evaluation (1) 3.7 (N/kg)	 (g =) $\frac{W}{m}$ allow numbers that round up to 3.7 (N/kg) award full marks for the correct answer without working	(2)

Q4.

Question number	Answer	Additional guidance	Mark
	substitution (1) (W=) 6.0 x 1.6 evaluation (1) 9.6 unit (independent mark) (1) newton	 award full marks for correct answer without working. accept N, n and/or Newton 9.6 N/n gains full marks	**(3)** **AO2** (2) **AO1** (1)

Q5.

	Answer	Additional guidance	Mark
(i)	substitution (1) 12 x 10 evaluation (weight =) 120 (N)		**(2)** **AO2**

	Answer	Additional guidance	Mark
(ii)	rearrangement (1) $(g) = \frac{W}{m}$ substitution (1) $\frac{20}{12}$ evaluation (1) (g on Moon =)1.7 (N/kg)	rearrangement and substitution in either order accept answers that round to 1.7 e.g. 1.67 (N/kg) award full marks for the correct answer without working 1.6 scores 2 marks 240 scores 1 mark 0.6 scores 1 mark 20 × 12 scores 1 mark $\frac{12}{20}$ scores 1 mark	(3) AO3

Q6.

Question number	Answer	Additional guidance	Mark
	substitution (1) (v^2 -0 =) 2x 10 x 1.5 evaluation (1) 5.5(m/s)	 accept numbers that round to 5.5 e.g. 5.477 30(m/s) gains 1 mark for correct substitution but no square root taken award full marks for correct answer without working.	**(2)** **AO2**

Q7.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) (t =) $\frac{10 - 6.2}{2.5}$ evaluation (1) (t =) 1.5 (s)	 $\frac{3.8}{2.5}$ allow $\frac{6.2 - 10}{2.5}$ or $\frac{-3.8}{2.5}$ 1.52 (s) allow –1.5(2) (s) award full marks for correct answer without working	**2** **AO2.1**

Question number	Answer	Additional guidance	Mark
(ii)	substitution OR rearrangement (1) $(-)10^2 = 2 \times (-)\ 4.4 \times x$ evaluation (1) (x =) 11 (m)	 $(x) = \frac{v^2 - u^2}{2 \times a}$ $(x=)\ \frac{(-)10^2}{2 \times (-)\ 4.4}$ allow values that round to 11 (m) e.g. 11.36 (m) ignore negative sign in answer line	**2** **AO2.1**
		accept 1.1(36) for one mark award full marks for correct answer without working	

Q8.

Question Number	Answer	Additional guidance	Mark
(i)	rearrangement (1) (m=) $\frac{F}{a}$ substitution (1) (m=) $\frac{5500}{2.2}$ evaluation (1) 2500(kg)	Rearrangement and substitution may be in either order (m=) $\frac{5.5}{2.2}$ ignore unit conversion until evaluation POT (power of ten) error, e.g. 2.5 (kg) award 2 marks award full marks for correct numerical answer without working If no other marks scored, award one mark for a clear conversion of units	(3)

Question Number	Answer	Additional guidance	Mark
(ii)	correctly identifies data points from the graph to calculate areas (1) use of distance = area under graph (1) substitution (1) (distance) = $\frac{(23-16)\times 13}{2}$ evaluation <u>rounded to 2sf</u> (1) 46 (m)	both 7 and 13 seen (distance) = $\frac{7\times 13}{2}$ (= 45.5) accept 45 (m) award full marks for correct numerical answer without working 91 scores 2 marks (incorrect area calculation using distance x time)	(4)

Q9.

Question Number	Answer	Additional guidance	Mark
	substitution (1) 1800 x 1.2 evaluation (1) 2200 (N)	accept 1800 kg x 1.2 m/s^2 reject 1800 x 1.2^2 2160 award full marks for the correct answer without working allow 1 mark total for 2200 OR 2160 with any other power of ten	**(2)**

Q10.

Question Number	Answer	Additional guidance	Mark
	substitution (1) (F=) 1200 x 2.4 evaluation (1) 2900 (N)	 accept 2880 (N) award one mark for power of ten error award full marks for the correct answer with no working	**(2)** **AO2**